HW 3 Name: Derek Devine, Tessa Lawler, Trevor Newton

1.

a) Explain why this network is acyclic (i.e. contains no cycle). *The cascade model is a simple toy model to help us think about energy flow in directed acyclic graph such as food webs and some power grids. Imagine N nodes with indices i = 1, . . . , N and place an undirected edge between each unique pair with probability p just as in the Erdos-Renyi random graph seen in class. Now add a direction to every edge such that each edge goes from node i to node i ¡ i. The index i then informs us of the position of a node in a hierarchy (e.g. trophic level in food webs, or distance from source in power grids) and the model can be used to help us understand the distribution of energy flow through the hierarchy.*

a) *Explain why this network is acyclic (i.e. contains no cycle).*

In order for a cycle to exist there must be a possibility of having a directed edge from i' to i where i' ¡ 1.

However per the problem statement i' ¿ i, so this is not possible by definition.

Also, for a cycle to exist by the definition of the problem, then there would have to be allowed more the $\binom{n}{2}$ *edges to exist.*

Specifically $2\binom{n}{2}$ *edges could exist if cycles were allowed, obeying the following, where* E_T *is the number of edges:* $\binom{n}{2}$ $E_T \leq 2\binom{n}{2}$

Following the arrows in the graph below from edges in to out also illustrates there are no cycles possible.

b) *What is the average in-degree (edges going in) of vertex i? What is the average out-degree (edges going out) of vertex i?*

Both the average degree in and the average degree out can be found by summing over the example degrees above and dividing by n. In the example, the sum is 6 for both in and out degree number of edges, and divided by 4, the answer is 1.5. So $N_E = \binom{n}{2n} p$ *where the numerator is the number of directed edges in a network of size n, and denominator, the total number of nodes n.*

Which simplifies to: $N_E = \frac{n-1}{2} p$

c) *Show that the expected number of edges that run from nodes i ¿ i to nodes i i is (ni i 2)p.*

Please see the image below which follows this explanation:

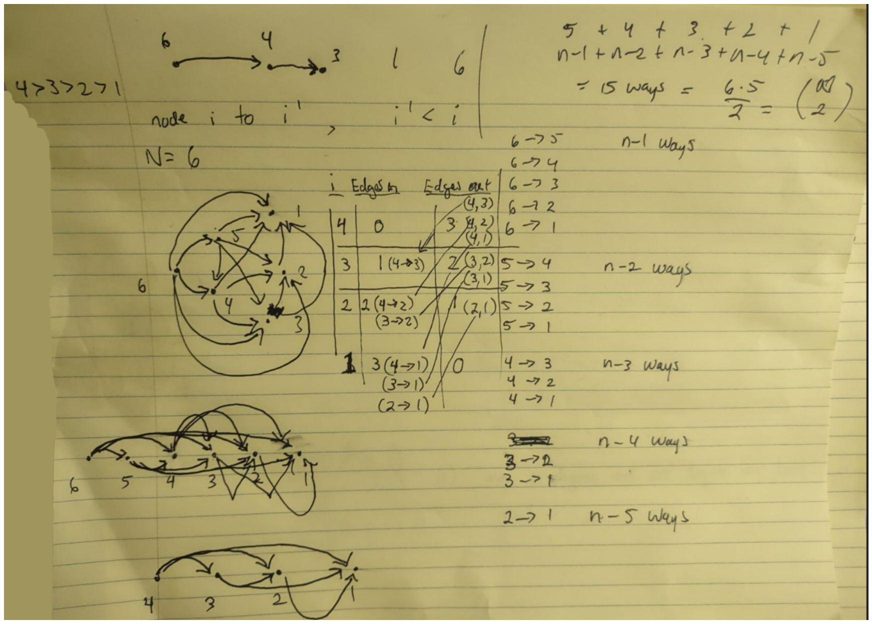


Figure 1: Shows why there are no cycles

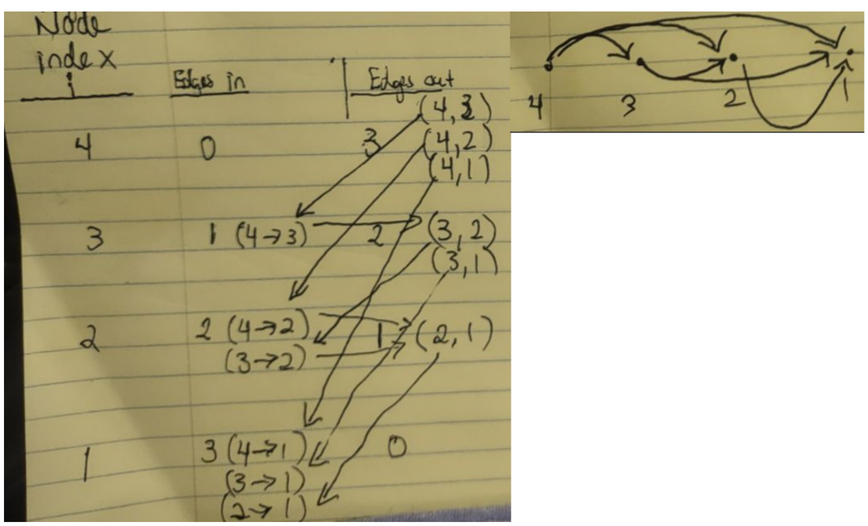


Figure 2: Some more visualizations of the directed nature of the graph

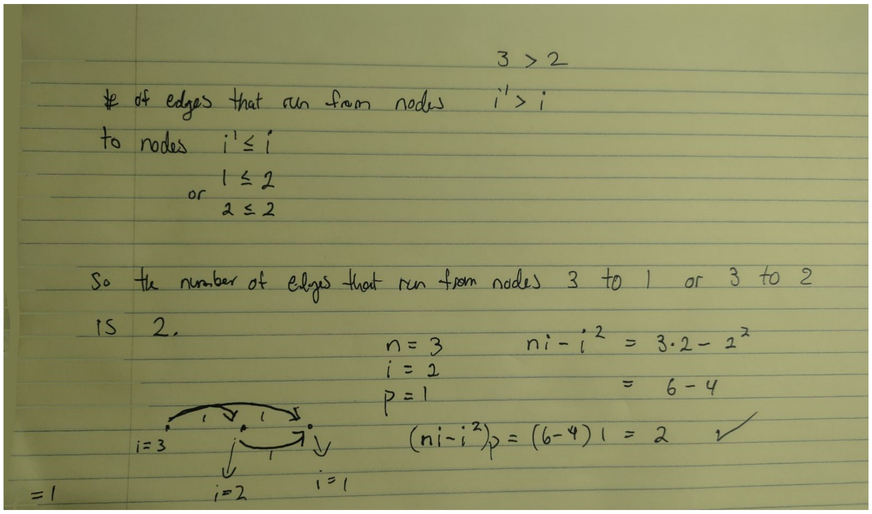


Figure 3: Logic proving the formula

The question asks for a formula for the number of edges that runs from nodes i' ¿ i to nodes i' i

Which means that if i is 2, as it is in the illustration below, then it's asking for the number of edges from a node index greater than 2 to 2, and from a node index less than or equal to 2 to the node index 1.

In other words it's asking for: The # of edges that run from node: i' ¿ i or in this example 3 ¿ 2, to nodes i' i , or in this case where 2 2 or where 1 2 .

So for this setup: n = 3 i = 2 p = 1 (the probability of each edge is the same, the graph is unweighted)

In plain english with this setup the problem is asking for the number of edges that run from node 3 to 2, or from node 3 to 1.

We see from the graph that there are N_E = 2 edges that satisfy the rules of the problem setup.

Next plugging in the values above into the formula also yields 2:

$(ni - i^2)p = 3(2) - 2^2 = 6 - 4 = 2$

Which verifies that the formula is accurate.

d) *Assuming N is even, what are the largest and smallest values of the quantity calculated in c) and where do they occur (in i)*

With n=3: *n=3, i = 2*

3 to 2 3 to 1 not 2 to 1 (2 isn't greater than 2)

n=3, i = 1

3 to 1 2 to 1 not 1 to 1 (because 1 isn't greater than 1)

With n=4:

n= 4, i = 3

4 to 3 4 to 2 4 to 1 not 3 to 2 (because 3 isn't greater than 3) not 3 to 1 (because 3 isn't greater than 3)

n= 4, i = 2

4 to 2 4 to 1 3 to 2 3 to 1 not 4 to 3 (because i=2, not 3) not 2 to 1 (because 2 isn't greater than 2)

n= 4, i = 1

4 to 1 3 to 1 2 to 1 not 1 to 1 (because 1 isn't greater than 1)

Summary:

We see that i is minimized at: **i=1, and i = n-1** *And that i i maximized at:* **i =**$\frac{n}{2}$

2. *Epidemic spreading on the configuration model allows us to test simple targeted vaccination strategies since not all individuals (nodes) are considered equal. Let us consider a simple example on an endemic Influenza epidemic, and test the impact of vaccination strategies knowing that around 40% of adults tend to get vaccinated.*

a) *Write a heterogeneous mean-field system for an SIS epidemic model on a configuration model network. You should obtain a system of equations for ˙I1, . . . , ˙Ikmax , where kmax is the largest degree in the network. Feel free to nondimensionalize your system if you wish.*

$\frac{d}{dt}N_I$ = -αN_I + KN_S $\frac{N_I}{N}$

I = $\frac{N_I}{N}$

$\frac{d}{dt}I$= -αI +KSI

$\frac{d}{dt}$ I_i $= -\alpha I_i$ + $k_i \frac{\sum_j A_{ij}[S_i I_j]}{k_i}$

$\frac{d}{dt}$ I_i $= -\alpha I_i$ + $S_i k_i \frac{\sum_j A_{ij}[S_i|I_j]}{k_i}$

$f_i = k_i \frac{\sum_j A_{ij}[S_i|I_j]}{k_i}$

$\frac{d}{dt}\ I_i = -\alpha I_i\ + (1 - I_i)f_i$

b) *Modify the above system of equations by adding new compartments for vaccinated nodes. Assume that*

$\frac{d}{dt}N_I$ = -αN_I + *(1-ρ)K*N_S $\frac{N_I}{N}$

I = $\frac{N_I}{N}$

$\frac{d}{dt}I$= -α*I* + *(1-ρ)KSI*

$\frac{d}{dt}$ I_i $= -\alpha I_i$ + *(1-ρ)*$k_i \frac{\sum_j A_{ij}[S_i I_j]}{k_i}$

$\frac{d}{dt}$ I_i $= -\alpha I_i$ + *(1-ρ)*$S_i k_i \frac{\sum_j A_{ij}[I_j|S_i]}{k_i}$

$\frac{d}{dt}$ I_i $= -\alpha I_i$ + *(1-ρ)*$(1 - I_i) f_i$

c) *Using the integrator of your choice, test how strong needs to be to eradicate an outbreak with the following parameters: transmission rate = 0.3 (week1) and recovery rate = 1 (week1) in a population with a geometric degree distribution pk = p(1 p) k with p = 1/4. Assume that a random 40% of the population gets vaccinated.*

Looking at the output below, running the simulation 4 different times over ten different rho values we can see that the minimum ρ value needed to completely eradicate the disease over all time periods appears to be somewhere around the .3 to .4 value. In other words it seems to be between p and the vaccination rate. I believe this suggests that in a fully connected homogeneous network, the value needed to eradicate the disease should be equal to the vaccination rate, in this case .4.

Also examining the bottom right hand image, we see that eradication was not achieved with the three lowest ρ values.

There also may be a connection with $1 - \rho$ and the and the $\alpha = \frac{2}{3}$ scaling exponent of a power law distribution in 2D, related to percolation.

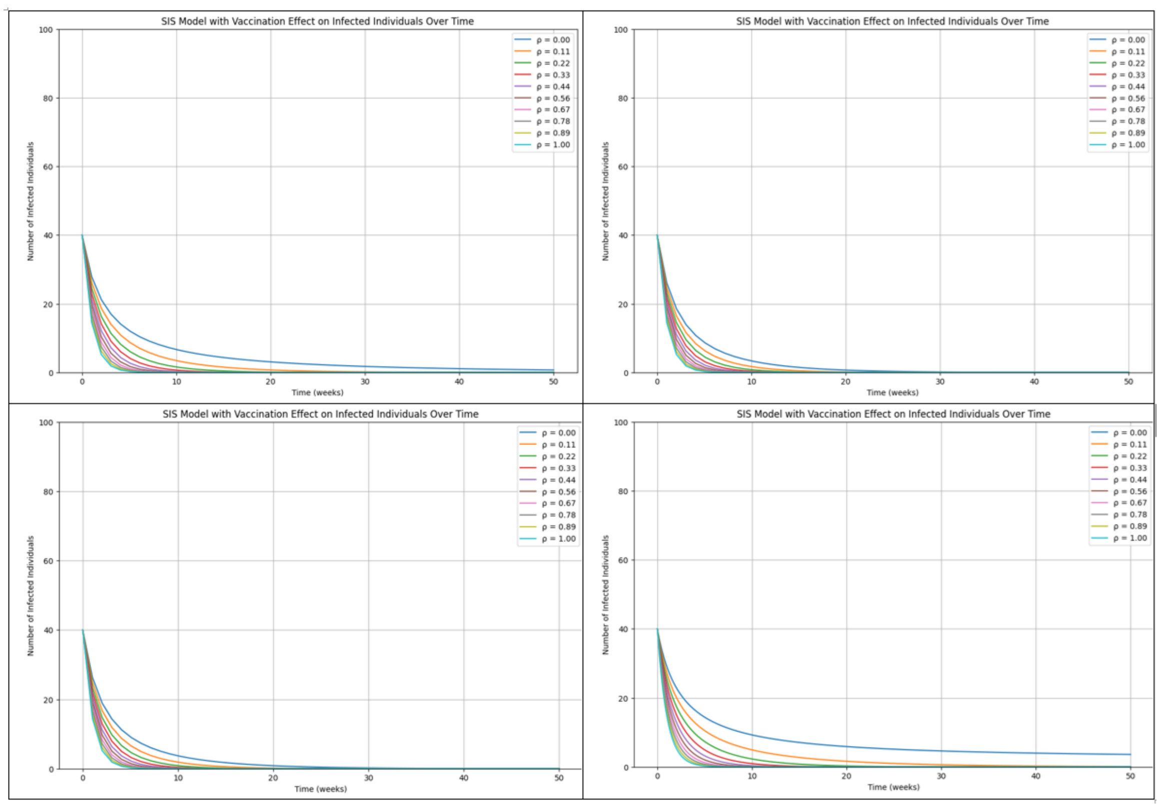


Figure 4: Four trials over ten values of rho.

d) *Repeat c) but now assume that the vaccination campaign convinced the 40% of*

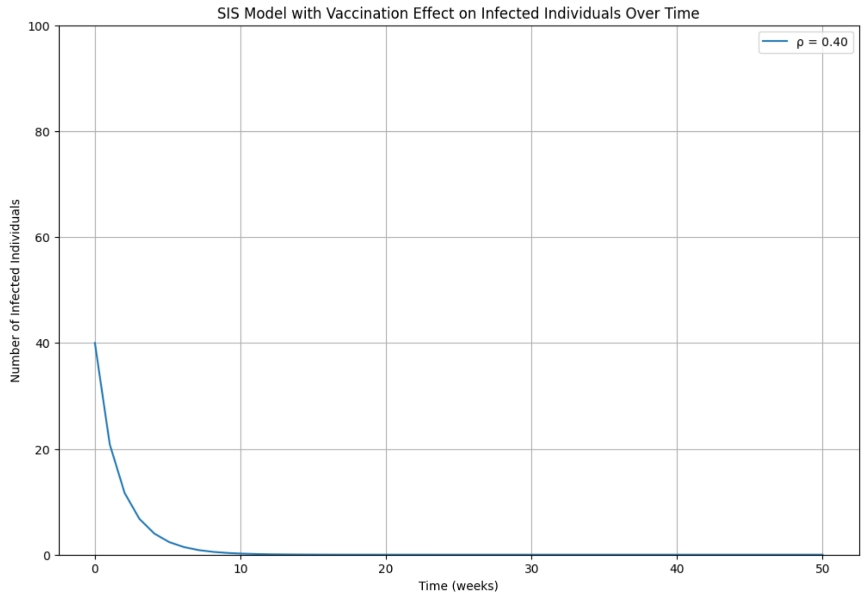


Figure 5: Only examining a rho value of .4, shows that it is completely eradicated

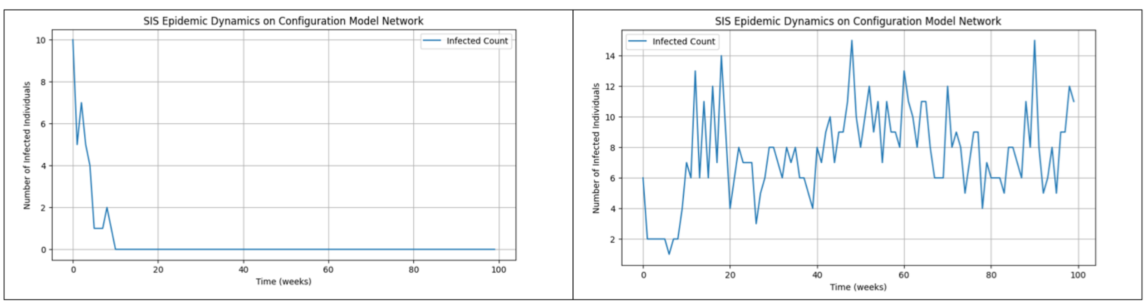


Figure 6: Rho of .4 on the left and .1 on the right

nodes with highest degree to get vaccinated.

At first glance, vaccinating only the highest 40% seems to indicate that the disease will spread easier, as the resulting network is more homogeneous after vaccinating the top 40% degree nodes. However, in addition to making the system more homogeneous, this vaccination scheme also has the effect of decreasing the average degree k of the network, effectively removing links/edges in the network, which leads to more disconnectedness. Ultimately the net effect is that it raises the epidemic threshold, suggesting that a lower ρ *value would be needed (than the case in 2c), in order to completely eradicate the disease.*

Upon performing a simulation, the result is below. We can see that vaccinating the top 40 percent, results in any rho value completely eradicating the disease, which is *consistent with the above intuition. Also of note the average degree k dropped from 4.28 in one of the above simulations in part c, to 3.48 after vaccinating the top 40*

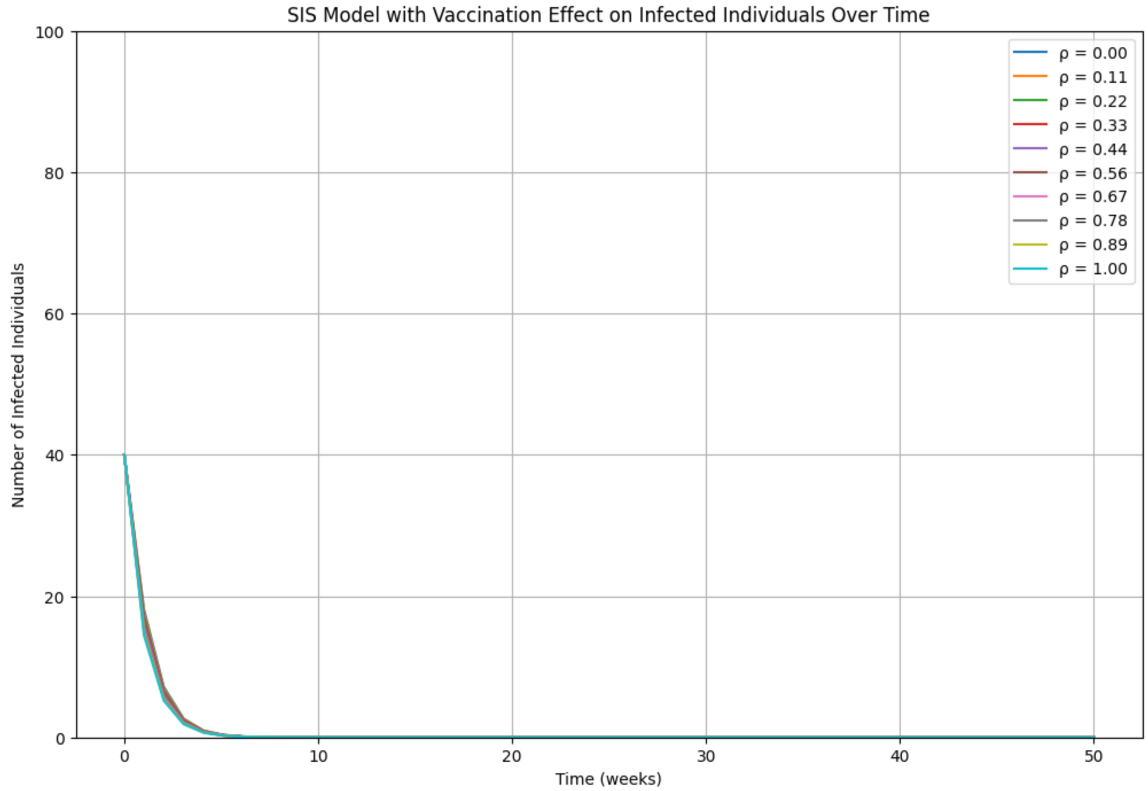


Figure 7: After vaccinating the top 40 percent, now all rho values eradicate the disease

3. *. Pick a network of your choice on Netzschleuder. You can easily import the network in Python via graph-tool (explained in the Netzschleuder webpage). Play around with network visualizations, share one of them in your write-up and note of what information if any you can glean from visualizing the network. Describe the network: What are the nodes? What are the edges? What is the average degree? What are some other static network measures, and why are they significant to this system? Tip: Avoid taking a network larger than a thousand of nodes to help you with the visualization and the next problem.*

An examination of the "bison" network from Netzschleuder: *this network depicts dominance in bison bulls during a few weeks of mating season in 1972. Each of the 26 nodes represents a bull. The edges between nodes represent dominance, operationalized as the outcome of a single incidence of mating-related aggression between bulls (i.e. the direction of the edges indicates the outcome of the "fight," not who initiated it.) The average degree of the network is 12.077. With only 26 nodes, this high average degree* *indicates the graph is dense, indicating that there's a lot of tension and competition between males during mating season.*

The visualization of this network demonstrates the variance in "connectedness" of bulls resulting from dominance behaviors during mating season. The network shows an extremely dense "knot" of nodes in the center, with some less-connected nodes on the fringes. This indicates that some bulls are more active participants in dominance displays than others. This is also supported by the relatively high degree standard deviation, which is 8.365. Some of the bulls on the exterior appear to have a significantly higher in-degree than out-degree, or vice versa, which may shed light on their positioning in the hierarchy of this population. Running parallel to this observation is the ratio of edge reciprocity, which is 0.586. This tells us that there is a significant amount of reciprocal dominance being displayed. In other words, conflicts between the same bulls frequently produce opposite outcomes, whereby Bull A may dominate Bull B in one interaction, but Bull B may in turn dominate Bull A in another. These statistics are helpful, because it is difficult to understand the behavior of the nodes in the "knot" in the center of the graph because the density of edges there obscures directionality.

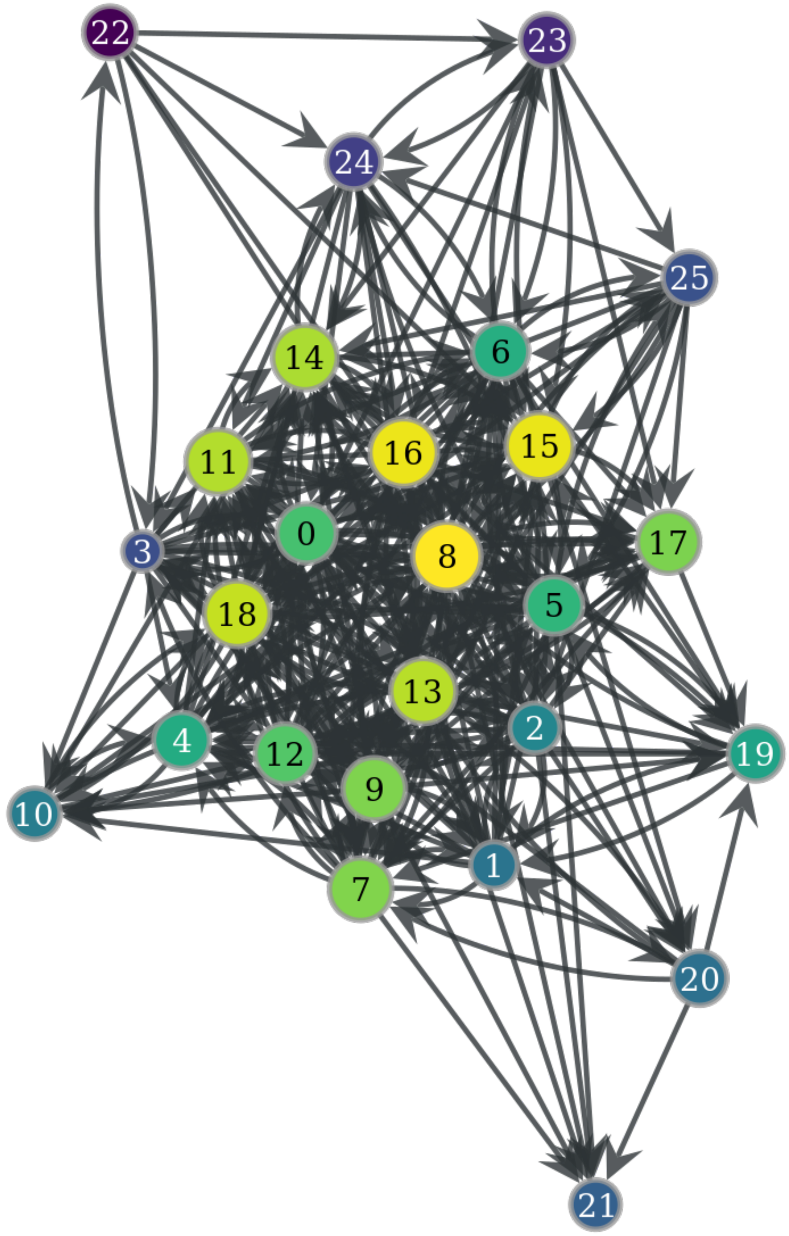


Figure 8: Bison Network Visualization

4. The voter model is the simplest computational analogy to social debate and allows us to study the ability of a network to reach a consensus. Nodes are typically in one of two states (say red and blue), and at every time step, a randomly chosen node adopts the state of one of its neighbors, also chosen randomly.

A Generic Voter Model run on an Erdos-Renyi random graph:

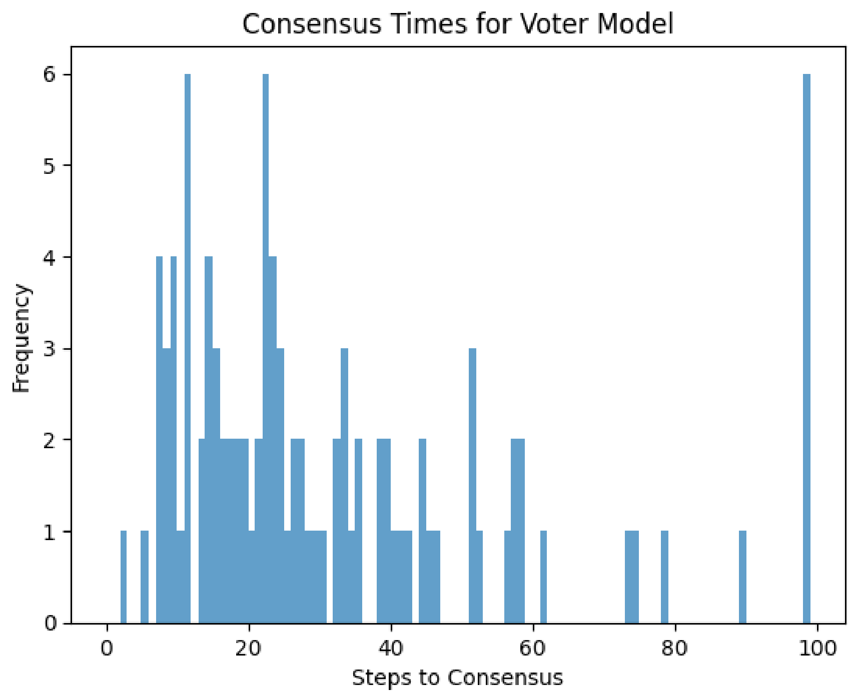


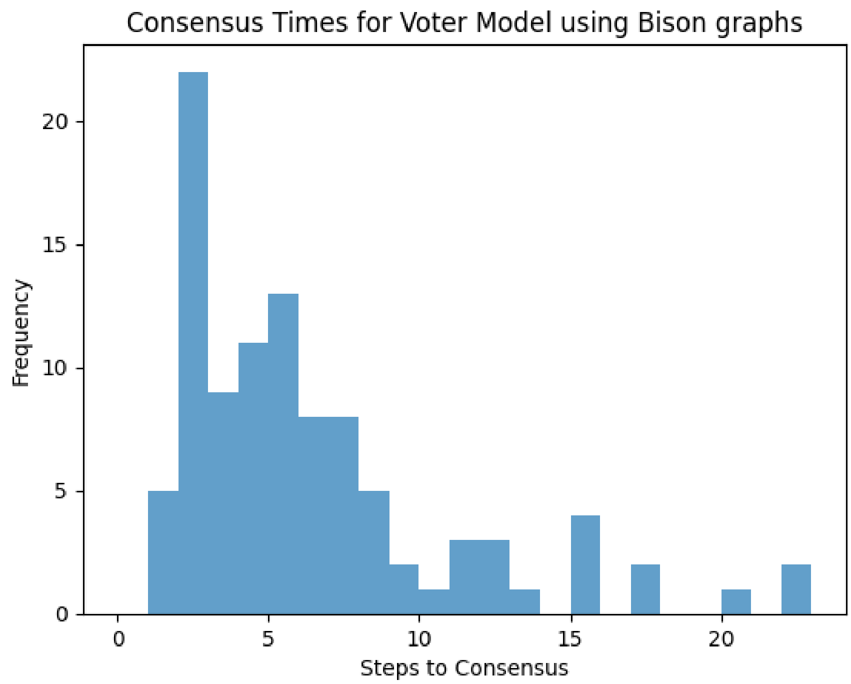
Figure 9: Voter Model run on an Erdos-Renyi random graph

```
mean 31.97
median 23.5
mode 99
stdev 24.560975451954572
variance 603.2415151515152
```

Figure 10: Statisstics for Voter Model run on an Erdos-Renyi random graph

a) *Using the network you chose in the previous problem, implement the voter model and run it on your network using some initial conditions (e.g., 60% red and 40% blue randomly distributed). Averaged over a few runs, does the network tend reach consensus? If so, how fast? Try a few different initial conditions.*

For the Bison Network: *consensus is reached after approximately 5-6 time steps, on average (mean = 5.87, standard deviation = 4.4).*



b) *Imagine running the same model on a version of your network randomized with the configuration model (or actually do it). How do you expect the dynamics to compare to the real network? Explain with in words or simulation results*

The dynamics will be influenced by whether the network configuration used exhibits greater heterogeneity compared to the Bison Social network described in question 3.

I believe there would be greater heterogeneity in the configuration model, particularly if the nodes followed a geometric degree distribution, which would lead to a larger average k. In this scenario, if vaccination were applied randomly rather than concentrating on just the top 40% of high-degree nodes, the network would exhibit increased clustering and would be more resilient to the spread of disease, which will lead to a slower consensus process.